Physica A 633 (2024) 129414



Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa





Anisotropic contractile particle model with avoidance for simulating pedestrian navigation in dilute and dense systems

Rafael F. Martin [a,*], Daniel R. Parisi [b]

[a] Instituto Tecnológico de Buenos Aires, Lavardén 315, C1437ETC, C. A. de Buenos Aires, Argentina
[b] Instituto Tecnológico de Buenos Aires. CONICET, Lavardén 315, C1437ETC, C. A. de Buenos Aires, Argentina

ARTICLE INFO

Keywords:
Pedestrian dynamics
Collision avoidance
Navigation
Steering
Off-lattice cellular automaton

ABSTRACT

We introduce a novel rule-based pedestrian simulation model that extends an existing one by incorporating enhanced particle contact rules and evasive maneuver capabilities for collision avoidance. The model is calibrated using individual trajectory data from avoidance experiments. With the same set of parameters, the new model satisfactorily replicates experimental observations, including the avoidance behavior of individual agents, fundamental diagrams of unidirectional and bidirectional flows, as well as specific flow rates through bottlenecks.

1. Introduction

The mechanisms of pedestrian navigation and collision avoidance are crucial in the dynamics of pedestrian movement in confined and populated spaces. It is essential to have improved and experimentally validated models of such behavior for simulation software to be useful in the design of pedestrian facilities, crowd management, urban planning, and safety.

There are several types of navigation and collision avoidance models. One of the first ones is the Social Force Model (SFM), which uses a repulsion force between particles to keep them from making contact [1,2]. However, while it is a way of avoiding collision, it could also present some side effects [3–5]. The Predictive Collision Avoidance Model [5] proposes an improvement by computing a different direction of the social repulsion based on future positions, thus producing more natural avoidance trajectories. In contrast, Wang et al. [6] propose a new mechanism that removes the social force term and instead adjusts the magnitude and direction of the desired velocity to achieve navigation.

There are also other types of navigation models based on heuristics [7,8] where future movement is determined based on a fixed set of rules or a series of nested discrete decisions. In velocity-based models, pedestrian dynamics are defined by determining future velocity based on environmental elements [9,10]. Additionally, there are data-driven models [11,12] that utilize machine learning, neural networks, or analysis of experimental data to determine particle movement.

In a previous work [13], we introduced a comprehensive framework for simulating pedestrian navigation. In this framework, simulated particles execute avoidance maneuvers exclusively by modifying their desired velocity, which is contingent on the environment, including its geometry and the presence of other pedestrians. This premise is grounded in the belief that avoidance behaviors in real-world scenarios can be effectively generated through the self-propulsion mechanism, typically represented by desired velocity. It is worth noting that this reference framework is independent of the type of low-level model used, whether rule-based, force-based, or otherwise.

Using this approach, we rely on the contractile particle model (CPM) [14] to equip it with an avoidance mechanism that dynamically reorients its desired velocity. The rule-based pedestrian model proposed here not only successfully replicates fundamental

* Corresponding author.
E-mail addresses: ramartin@itba.edu.ar (R.F. Martin), dparisi@itba.edu.ar (D.R. Parisi).

https://doi.org/10.1016/j.physa.2023.129414
Received 28 April 2023; Received in revised form 22 November 2023
Available online 30 November 2023
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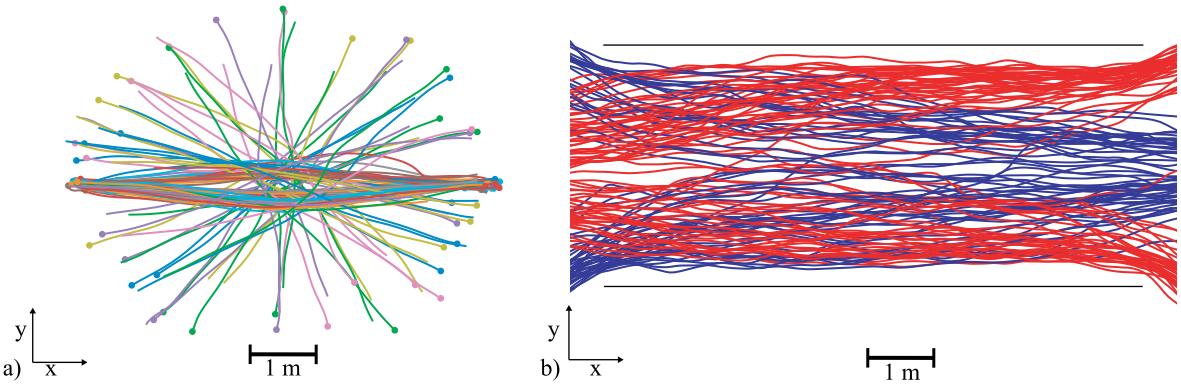


Fig. 1. Experimental trajectories. (a) MOCAP experimental trajectories for two colliding pedestrians [12]. (b) Forschungszentrum Jülich corridor bidirectional experimental trajectories [15].

diagrams of unidirectional flows and specific flow rates through exits, like the CPM [14], but it also has the ability to reproduce experimental observables from bidirectional flow and avoidance scenarios in dilute conditions (collisions between two pedestrians and between a pedestrian and a fixed obstacle). All of this phenomenology is captured using the same model and the same set of parameters.

2. Experimental data

We utilized data from our own experiments for one and two avoiding pedestrians, and from a public database from Forschungszentrum Jülich for more crowded scenarios.

The dilute scenario consisted of a collection of experimental trajectories obtained at the Motion Capture (MOCAP) Laboratory at the Instituto Tecnológico de Buenos Aires (ITBA) [12]. The data are collected by instructing each pair of volunteers to stand at different positions on a circle with a radius of 3 m and walk to the opposite side while avoiding the other volunteer in a natural manner. The resulting 40 pairs of trajectories are displayed in Fig. 1(a). We also considered another set of 10 experimental trajectories used in [12] in which individual volunteers were instructed to walk from one point of the same circle to the opposite side while avoiding a fixed obstacle of approximately 0.3 m width at the center.

The crowded scenario consisted of experimental trajectories published by the Forschungszentrum Jülich [15,16]. Specifically, we analyzed the trajectories associated with a corridor of length 8 m and two different widths, 3 m and 3.6 m. We considered data corresponding to bidirectional flow with the same flow rate in each direction. Fig. 1(b) shows the set of trajectories obtained in the case of the lower input flow in both directions, and 3.6 m width.

3. Models

In this study, we discuss three distinct models: the Contractile Particle Model (CPM) proposed by Baglietto and Parisi [14] that we consider as the base model, and two modifications taking into account anisotropy and avoidance.

The original CPM proposes that individual particles have a variable personal space radius. Whenever another particle enters this personal space, the velocity of the initial particle reduces to zero only during one simulation time step, and the personal space radius becomes equal to the particle's physical and impenetrable radius. This reaction occurs independently of the direction through which the invading particle enters the personal space, making the model isotropic. While this microscopic mechanism produces good results at a macroscopic scale in unidirectional systems, it is not appropriate for bidirectional systems in which two near and parallel-moving particles are unable to approach each other without triggering a reaction. To address this issue, we present a second model called Anisotropic Contractile Particle Model (A-CPM), in which particles only react to personal space invasions that occur at the front of their physical radius, which presents a narrower cross section than the personal space given by the current radius. The third variant is a modification of A-CPM that enables particles to anticipate collisions and avoid them by performing evasive maneuvers. We call this model Avoidance Anisotropic CPM (AA-CPM). In subsequent sections, we provide a detailed description of these models.

3.1. CPM

The CPM [14] is an off-lattice cellular automaton, with particles occupying continuous positions in 2D space and rules for updating their velocities.

Each particle i is defined by its position $\mathbf{r}_i$, a personal space radius $r_i \in [r_{min}, r_{max}]$, and a desired velocity $\mathbf{v}^d$ pointing towards a given target $\mathbf{T}_i$. The particle's radius should only be interpreted as a hard physical radius when it is at its minimum value (r_{min}). But for larger radii, it represents a personal space needed for walking. The magnitude of the particle's velocity is a function of r_i. When the personal space radius is at its maximum, the particle moves at the desired speed v^d_{max}.

In the original Contractile Particle Model, two particles i and j are considered to be interacting if their distance is less than the sum of their personal space radii, i.e., $|\mathbf{r}_i - \mathbf{r}_j| < r_i + r_j$. Similarly, a particle is in contact with a wall if its distance to the nearest point on the wall is less than r_i.

The basic structure of movement and collision dynamics in the CPM has been implemented with a trunk structure that remains similar across all three models. For each particle, at each time step, it is analyzed whether the particle is in contact with another particle or wall. If it is not in contact with other elements, the particle personal space radius is increased until reaching the maximum radius r_{max}, and the particle is moved towards the goal in the direction of $\mathbf{e}^t = \frac{\mathbf{r}_i - T_i}{|\mathbf{r}_i - T_i|}$. On the other hand, if it is in contact with other elements, the radius is reduced to the minimum value, r_{min} (the physical radius of the particle) (see Eq. (1)), and it is moved in the direction opposite to the contacting element ($\mathbf{e}^{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}$) at the desired speed v^d_{max} during just one simulation time step: collision with other particles is the sole mechanism that generates a reduction in the radius.

More formally, in each iteration the radius and position of the particles are updated according to Eqs. (1) and (2):

$$r(t) = \begin{cases} r_{min} & \text{if } i \text{ is in a collision} \\ r(t-\Delta t) + r_{max}.\Delta t/\tau & \text{if } i \text{ is not in a collision and } r(t-\Delta t) + r_{max}.\Delta t/\tau < r_{max} \\ r_{max} & \text{otherwise} \end{cases} \tag{1}$$

$$\mathbf{r}_i(t) = \mathbf{r}_i(t-\Delta t) + \mathbf{v}^d_i(t)\Delta t \tag{2}$$

with

$$\mathbf{v}^d_i = \begin{cases} v^i \mathbf{e}^t & \text{if } i \text{ is not in contact} \\ v^d_{max} \mathbf{e}^{ij} & \text{otherwise} \end{cases} \tag{3}$$

To define the magnitude of the velocity v_i in cases where the particle is not in contact with another particle, the formula in Eq. (4) is used, which relates it to the particle's radius r_i.

$$v_i = v^d_{max} \left(\frac{r_i - r_{min}}{r_{max} - r_{min}} \right)^{\beta} \tag{4}$$

The values of the parameters β, τ, r_{min} and r_{max} are in accordance with the original CPM [14] and are provided in Section 4.1.

3.2. A-CPM

While the original CPM satisfactorily reproduces observed data for unidirectional flows, it presents difficulties when attempting to simulate bidirectional flow because the cross-sectional area of the isotropic particle (for $r > r_{min}$) is higher than in real pedestrian systems, resulting in artificial collisions and decreased speed. To overcome this issue, we propose the anisotropic CPM (A-CPM), which has different contact detection criteria. Inspired by the work of Tordeux et al. [17], we defined that particles in the A-CPM collide only within a frontal margin, making it a more permissive and asymmetric approach.

We consider that a particle enters in contact with another if the personal space in front of the particle overlaps with the personal space of the other, as illustrated in Fig. 2(b). Formally, a particle i, with position $\mathbf{r}_i$ and velocity $\mathbf{v}_i$, is in contact with particle j, with position $\mathbf{r}_j$ and personal space radius r_j, if either of the following conditions holds, depending on r_i:

- $r_i = r_{min}$ and $|\mathbf{r}_i - \mathbf{r}j| < r_i + r_j$; or
- $r_i \neq r_{min}$ and the angle β formed between $\mathbf{v}_i$ and $\mathbf{r}_j - \mathbf{r}_i$ lies between $-\pi/2$ and $\pi/2$.
  Moreover, the following two conditions must be satisfied simultaneously:
  - The lines parallel to $\mathbf{v}_i$ from the endpoints to the circle of radius r_{min} intersect the circle centered at $\mathbf{r}_j$ with radius r_j.
  - The two particles overlap in their current radii, that is, $|\mathbf{r}_i - \mathbf{r}_j| < r_i + r_j$.

To illustrate these conditions, in Fig. 2(a) we can observe that particle h fulfills only the first condition, as indicated by the two marked black dots. Meanwhile, particle j satisfies solely the second condition due to the overlap. As a result, neither of these particles is considered to be colliding with particle i. On the other hand, in Fig. 2(b) we can see an example of a particle that, by simultaneously meeting both conditions, we consider to be colliding with particle i.

In the A-CPM, a particle collides with a wall only if the distance to the nearest point on the wall is less than its minimum radius r_{min}.

3.3. AA-CPM

Now, we are adding an avoidance layer to the previous model based on the general idea proposed in [13]. Evasive maneuvers are expressed through changes in the desired velocity, using temporal and dynamic targets.

The basic dynamics follow the same structure as described in Section 3.2, but now the desired velocity $\mathbf{v}^d_i$ used in Eq. (2) is modified by Eq. (5) while navigating by following the avoidance unit vector $\mathbf{e}^a$ computed by Eq. (6):

$$\mathbf{v}^d_i = \begin{cases} v^i \mathbf{e}^a & \text{if } i \text{ is not in frontal contact} \\ v^d_{max} \mathbf{e}^{ij} & \text{otherwise} \end{cases} \tag{5}$$

$$\mathbf{e}^a = ((\sum_{j=1}^{2} \mathbf{n}_c^j) + \mathbf{n}_c^w + \mathbf{e}^t)/|(\sum_{j=1}^{2} \mathbf{n}_c^j) + \mathbf{n}_c^w + \mathbf{e}^t|, \tag{6}$$

where the direction to the final target is $\mathbf{e}^t = \frac{\mathbf{T}_i - \mathbf{r}_i}{|\mathbf{T}_i - \mathbf{r}_i|}$, $\mathbf{n}_c^w$ is the collision vector with walls, and $\mathbf{n}_c^1, \mathbf{n}_c^2$ the collision vectors for each of the two particles under consideration (in our model we will use the two closest particles that are in front, but with a slight modification of Eq. (6) as many particles as desired can be considered). These vectors are calculated as follows (see Fig. 2(c) and (d)).

First, we need to calculate $\mathbf{e}_c^{ij}$ for each particle j located ahead of particle i, i.e., such that the angle between $\mathbf{v}_i$ and $(\mathbf{r}_j - \mathbf{r}_i)$ lies in $[-\pi/2, \pi/2]$:

- Calculate the relative velocity $\mathbf{v}_{ij} = \mathbf{v}_j - \mathbf{v}_i$.
- Compute the angle β between $\mathbf{v}_{ij}$ and the direction towards the target.
- Analyze the value of β.
  - If $|\beta| < \frac{\pi}{2}$, $\mathbf{e}_c^{ij} = (0, 0)$. This prevents maneuvers to avoid a particle that is moving away from it.
  - If $|\beta| \geq \frac{\pi}{2}$:
    - Construct the unit vector $\mathbf{e}^{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}j|}$.
    - Calculate the signed angle α between $\mathbf{e}^{ij}$ and the relative velocity $\mathbf{v}_{ij}$.
    - Compute $f(\alpha)$, where $f(\alpha) = ||\alpha| - \frac{\pi}{2}|$.
    - Rotate the unit vector $\mathbf{e}^{ij}$ by $-\text{sign}(\alpha).f(\alpha)$ radians to obtain $\mathbf{e}_c^{ij}$. We have used, as a convention, that positive angles are those counterclockwise.
- Calculate $\mathbf{n}_c^j = \mathbf{e}_c^{ij}\ A_p\ exp(-d_{ij}/B_p)$, where A_p and B_p are adjustable parameters, and d_{ij} is the distance between particle centers $\mathbf{r}_i$ and $\mathbf{r}_j$.

Fig. 2(c) shows an example of the elements necessary to form one of the collision vectors, the one related to particle j in this case.

The function f is specifically designed to avoid collisions by causing the particle to evade perpendicularly to $\mathbf{e}^{ij}$ when the potential collision is head-on. As the angle of the potential collision changes, the rotation of the vector $\mathbf{e}^{ij}$ becomes less severe.

Similarly, to avoid collisions with walls, the following steps are taken. First, the closest point $\mathbf{w}$ on a wall to the particle is detected. Then, the unit vector $\mathbf{e}^{iw} = \frac{\mathbf{w} - \mathbf{r}_i}{|\mathbf{w} - \mathbf{r}i|}$ is constructed. Finally, the avoidance vector $\mathbf{n}_c^w = \mathbf{e}^{iw}\ A_w\ exp(-d_{iw}/B_w)$ is calculated, where A_w and B_w are two adjustable parameters, and d_{iw} is the distance from the particle to point $\mathbf{w}$. In Fig. 2(d) we can see how by adding all these collision vectors (both those related to particles and the one corresponding to the wall) and the vector towards the goal, we obtain the unit vector $\mathbf{e}^a$.

We emphasize that this model is rule-based; nevertheless, the functional form used to weigh the avoiding interactions among different particles and obstacles was inspired by the social force term of the SFM [2].

4. Results

4.1. Calibrating the avoidance parameters

The parameters related to the basic CPM are taken directly from Ref. [14]. They are: $\beta = 0.9$, $\tau = 0.5$ s, $r_{min} = 0.15$ m, and $r_{max} = 0.32$ m. The simulation time step is defined as $\Delta t = 1/33$ s in order to match the frame rate of the experimental data taken in the MOCAP Laboratory at ITBA.

The new parameters introduced in the elusive layer by the AA-CPM, namely A_p and B_p, are estimated using the experimental trajectories from two colliding pedestrians (X_p^e) shown in Fig. 1(a) and also, from one pedestrian avoiding a fixed obstacle (X_o^e), as described in Section 2. The starting position and velocity of each experimental trajectory are used as the initial conditions for the corresponding particle simulation, with the final goal set to the last recorded point of the trajectory.

In Eq. (7) we define an error metric (E) that enables a comparison between simulated and experimental trajectories in all the scenarios mentioned above.

$$E(X^s, X^e) = \frac{1}{\#(X^s)} \sum_{\mathbf{r}_s \in X^s} \min_{\mathbf{r}_e \in X^e} (|\mathbf{r}_s - \mathbf{r}_e|), \tag{7}$$

where $X^s = X_p^s \cup X_o^s$ and $X^e = X_p^e \cup X_o^e$ are the sets of points in the simulated and experimental trajectories, respectively, and the subscript 'p' indicates pedestrian-pedestrian trajectories, and 'o' denotes pedestrian-obstacle trajectories.

This metric calculates the average distance from each point in the simulated trajectory to the nearest point in the experimental one. It is important to note that both trajectories should have the same starting and ending points for this metric to be meaningful.

In Fig. 3(a), the mean error $E = \langle E(X^s, X^e) \rangle$ for all the 90 trajectories is presented. The values at which collisions occurred between the physical radius of the particle and the fixed obstacle are the gray area.

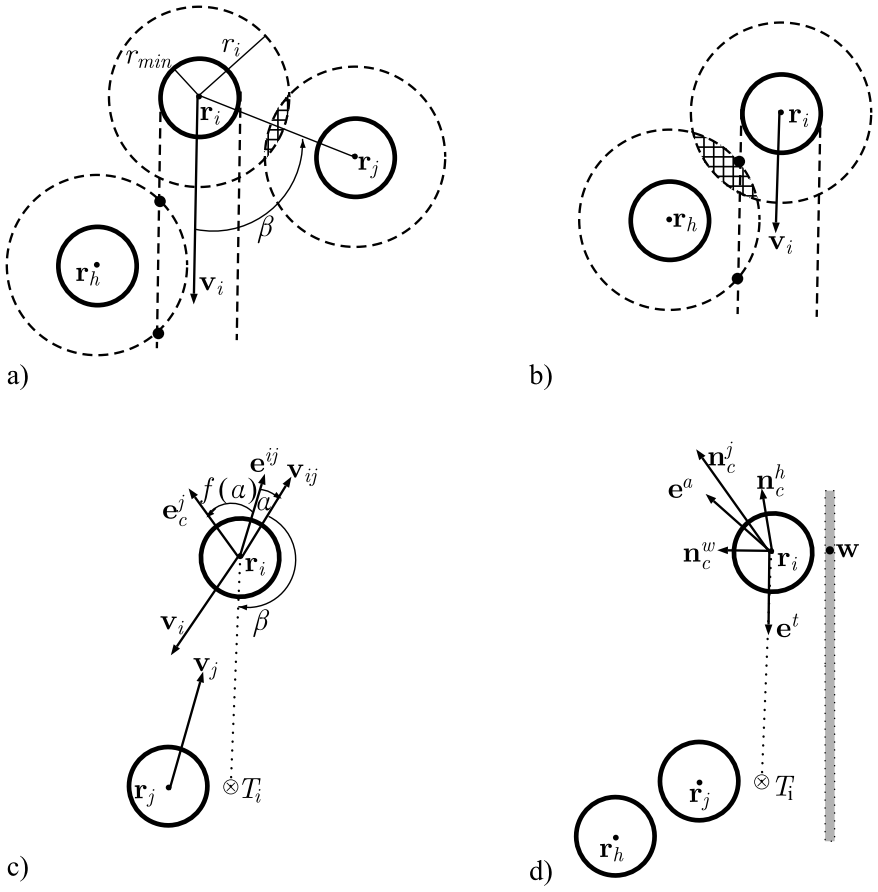


Fig. 2. Key elements for defining the CPM and the proposed modifications. (a) Example of two particles that are not in contact with i in A-CPM and AA-CPM. (b) Example of a particle that is in contact according to the new models. (c) Elements for collision vectors. (d) Example of vector sum to obtain $\mathbf{e}^a$.

Also, we compute the difference between the mean error considering only the 10 trajectories with fixed obstacle ($\langle E(X_o^s, X_o^e)\rangle$) and the mean error with the 40 pedestrian-pedestrian pairs of trajectories ($\langle E(X_p^s, X_p^e)\rangle$) according to Eq. (8).

$$\Delta E = |\langle E(X_o^s, X_o^e)\rangle - \langle E(X_p^s, X_p^e)\rangle| \tag{8}$$

A heat map of ΔE as a function of A_p and B_p is presented in Fig. 3(b). Based on Fig. 3(a) and (b), we decided to adopt the values $A_p = 1.1$ and $B_p = 2.1$. These values result in a low overall error of $E = 0.24$ m and exhibit nearly negligible differences in errors for both scenarios (ΔE), making them a more uniform parameter set for both cases. Examples of experimental trajectories and their corresponding simulations using these parameter values are shown in Fig. 3(c) and (d).

From this study with individual trajectories, we have already determined the avoidance parameter values between pedestrians. From here on, we will validate the AA-CPM model with these parameters in more complex systems that involve higher densities.

4.2. Unidirectional and bidirectional flow

In this section, we explore the corridor scenario from [18], where we deliberately selected the particle–wall avoidance parameters A_w and B_w through a qualitative process. Specifically, we set $A_w = 1215$ and $B_w = 0.025$. With this choice, when a particle grazes the wall, the magnitude of the collision vector is approximately $|\mathbf{n}_c^w| \sim 10$, while at a distance of 0.3 m from the wall (twice the minimum radius), the magnitude decreases to $|\mathbf{n}_c^w| \lesssim 0.1$. These values enable particles to prevent overlapping their hard physical cores (r_{min}) with the walls while also allowing them to make full use of the corridor width, thus ensuring that artificial distancing from the walls is not introduced.

The fully calibrated AA-CPM, using the parameters determined above, is tested by simulating both unidirectional and bidirectional flows in a corridor. The results are then compared with experimental data and with the other two models, serving as benchmarks.

In the unidirectional flow case, simulations are performed with a corridor having dimensions of 8 m in length and 3.6 m in width. Initially, N particles are randomly placed in the corridor, and the particles are given a starting speed of 0. The desired speed is

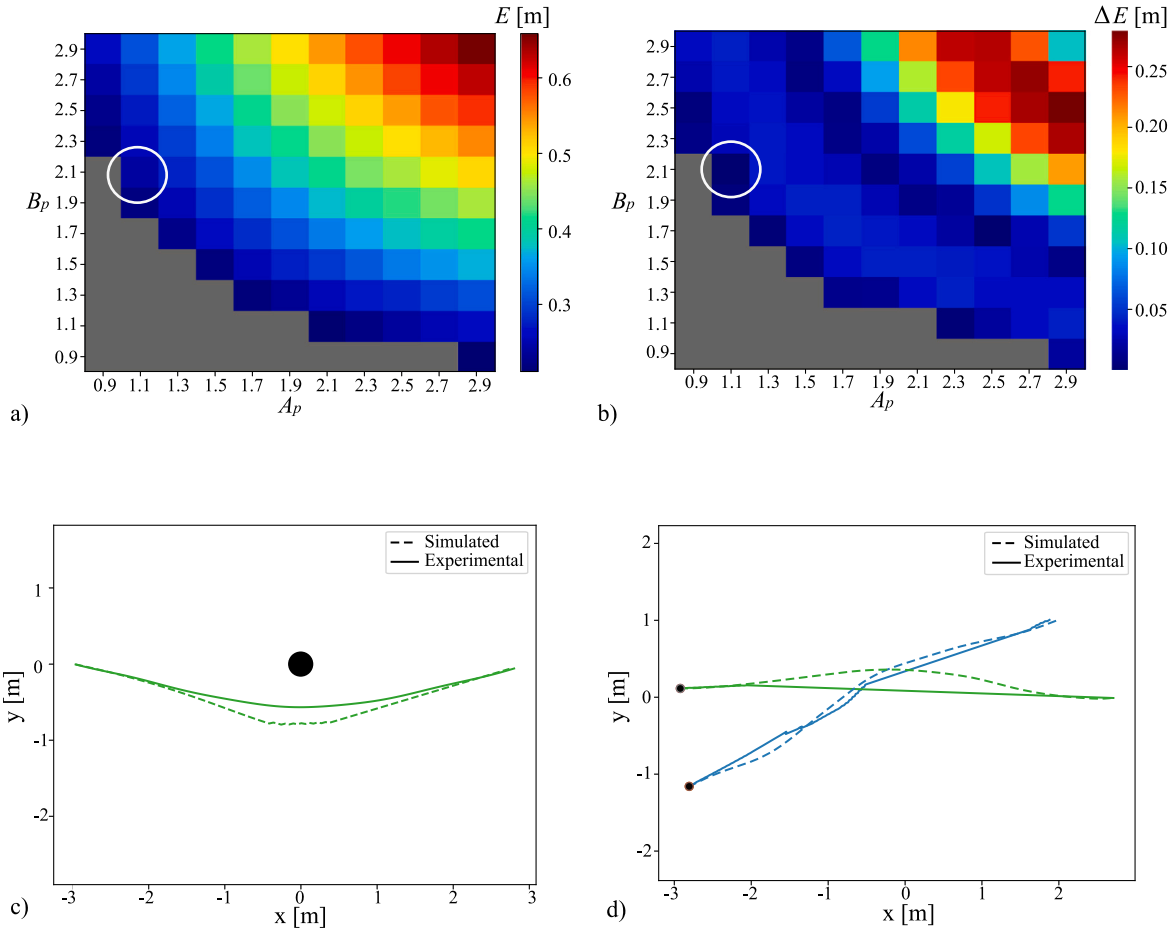


Fig. 3. Calibration of avoidance parameters. (a) Error defined by Eq. (5) considering all the experimental trajectories. (b) Absolute value of the difference between the mean error in pedestrian-obstacle cases and pedestrian-pedestrian cases (Eq. (8)). (c) Example of experimental and simulated trajectories avoiding a fixed obstacle with the chosen parameters: $A_p = 1.1$ and $B_p = 2.1$. (d) Example of two moving pedestrian trajectories, experimental and simulated ones, using the same chosen parameters.

randomly chosen in the range 1.55 ± 0.155 m/s, where 1.55 m/s is the reported average free walking speed in [18]. The number of particles varies between 3 and 90, thus covering the range of densities similar to the experiments.

The final target for each particle is at the end of the corridor, with the same vertical coordinate y as the particle (see Fig. 1(b)). The target is updated with every simulation time step, and periodic boundary conditions are implemented. The duration of each simulation is set to 30 s.

To evaluate the performance of the proposed model in comparison with the existing models and experimental data, we constructed a fundamental diagram. For this purpose and to mitigate fluctuations in instantaneous speed, we opted to calculate velocity considering the particle's displacement over $5\Delta t$, more formally, $v(t) = \frac{|\mathbf{r}(t)-\mathbf{r}(t-5\Delta t)|}{5\Delta t}$.

To determine the density, we use a nonparametric method described in Refs. [19,20] and consider a version of this method that deals with finite simulation domains [21]. The formula $\rho = (k-1)/A_k$ can be used to calculate the local density at any point in space, where k is the number of nearest neighbors. Specifically, we use the positions of each pedestrian as points for calculating the density, and the corresponding pedestrian is counted as one of the k neighbors. The area A_k is the intersection between the interior of the corridor and the circle centered at the particle with a radius equal to the distance from the particle to the kth nearest neighbor. We selected $k = 5$ based on the findings in Ref. [21], which reported a difference of less than 0.35% compared to the Voronoi cell method [22]. This procedure allows us to obtain points $(\rho_i(t_i), v_i(t_i))$ for all particles and time steps. From these microscopic data, we compute the fundamental diagram by moving average with $n = 10000$.

Fig. 4(e) presents curves corresponding to both simulated systems and experimental data, as reported in the bibliography [23,24]. It is evident that the AA-CPM exhibits a trend more similar to the experimental speed-density relationships. In contrast, the A-CPM, which employs a relaxed collision criterion, results in fewer reductions in particle speed compared to both CPM and AA-CPM. The lower speed observed in AA-CPM may be attributed to avoidance maneuvers, leading to subsequent lateral movements that do not allow achieving the maximum possible displacement.

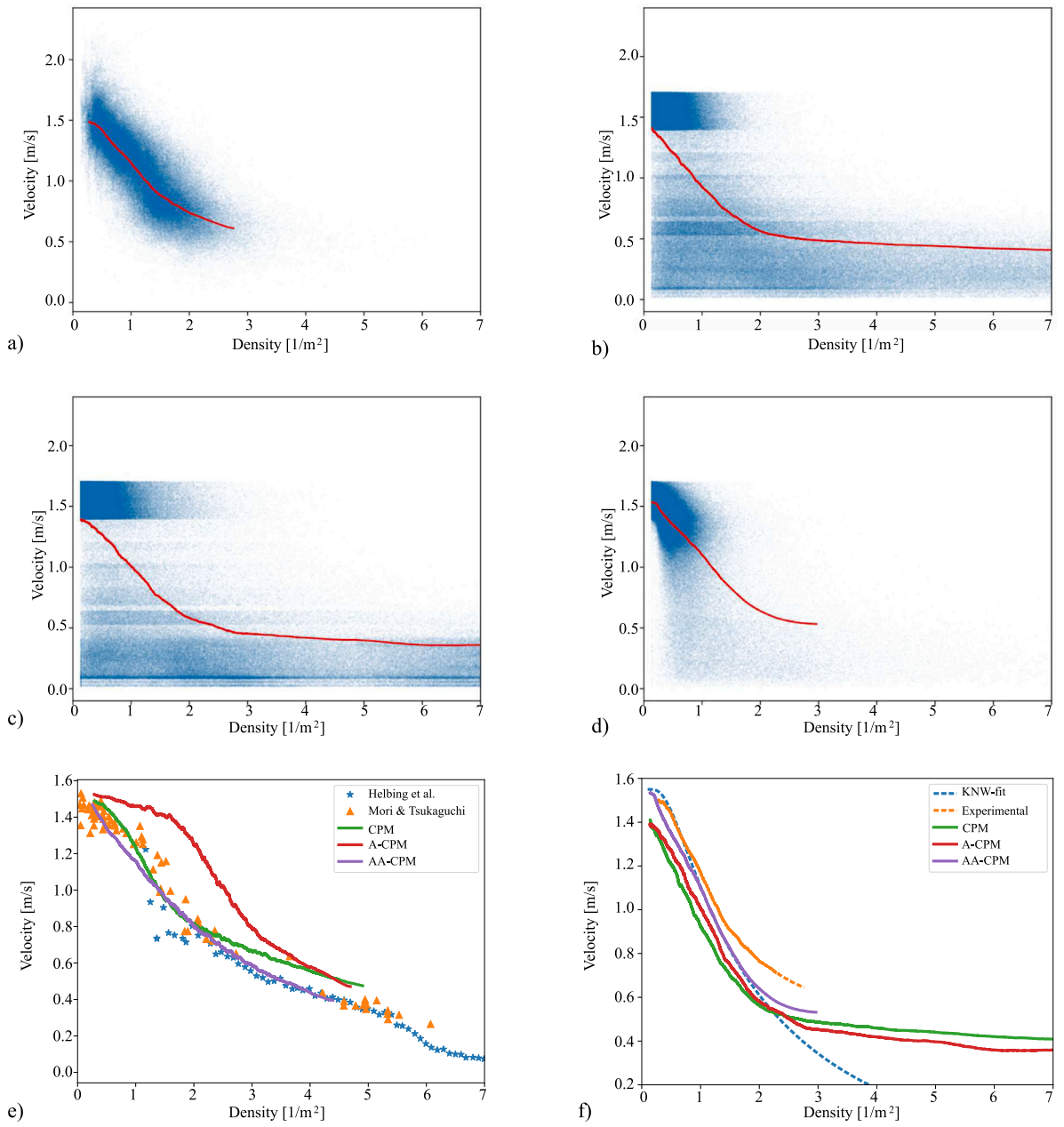


Fig. 4. Scatter plots and fundamental diagrams. (a) Experimental. (b) CPM. (c) A-CPM. (d) AA-CPM. (e) Unidirectional flow. (f) Bidirectional flow.

In order to investigate a more complex scenario, we consider the bidirectional flow within a single corridor, taking the experimental data from [15,16]. The input flow rate is symmetrical at both ends of the corridor, with a range of $Q_{in} \in [1.8, 3.8]$ 1/s.

The simulations have the following configuration: The corridor dimensions are 16 m in length and 3.6 m in width, $N = 100$ particles are introduced alternatively at both ends of the corridor at fixed flow rates that range from $1/(33\Delta t)$ to $1/(8\Delta t)$.

The initial position of the particles is randomized at a point $(0, Y)$ or $(16, Y)$, depending on the side, with Y uniformly distributed between 0.15 m and 3.45 m.

The desired velocity of the particles and final targets are set in the same way as in the unidirectional flow. Particles are removed from the system after traveling 14 m of the 16-meter-long corridor. The initial conditions for the three models are identical. Approximately 100 simulations are performed with distinct initial conditions for each input flow rate.

In Fig. 4(a) to (d), we can observe the scatter plot (ρ, v) corresponding to the experimental data [15,16], and the CPM, A-CPM, and AA-CPM models, along with the moving average fitting. Fig. 4(f) presents all the averaged fundamental diagrams obtained from

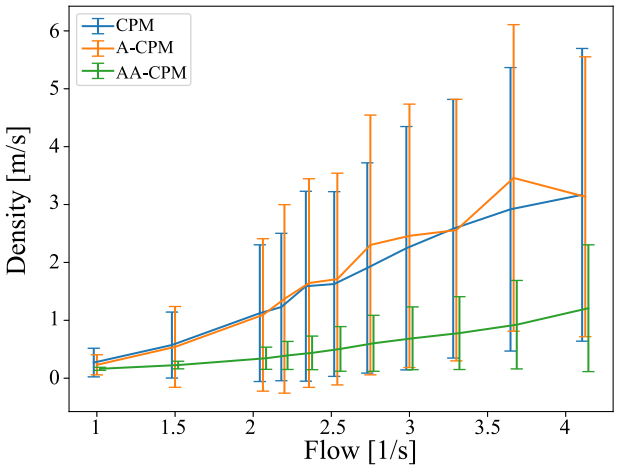


Fig. 5. Mean density obtained after simulations with different input flow values.

the bidirectional flow simulations and the Kladek–Newell–Weidmann formula [25] $v(\rho) = v_o(1-\exp(1/\rho - 1/\rho_{max}))$ with $v_0 = 1.55$ m/s, $g = -1.913$ /m^2 y $\rho_{max} = 6$ /m^2. Here we can observe how all three models remain close to the experimental data and the KNW fit, with AA-CPM being closer at lower densities values. The most notable difference, as confirmed by the scatter plots, is that both CPM and A-CPM generate data with significantly higher densities than those observed in the experiments. This is primarily attributed to the creation of artificial congestion resulting from the lack of avoidance capacity in these models. This is confirmed by Fig. 5, where we can observe that as the particle input flow rate increases, the first two models generate higher densities, whereas in the case of AA-CPM, the increase in density is less pronounced.

Next, we delve deeper into characterizing the artificial congestion. A relevant antecedent for similar systems was presented in the work of Qiancheng et al. [26], where they investigated local and global jamming by observing the average velocity of particles to determine their state of being stuck or not.

In our case, we employ a cluster analysis. We define a cluster at time t as a set C_t such that:

$$C_t : i \in C_t \iff \exists j \in C_t / |\mathbf{r}_i(t) - \mathbf{r}_j(t)| < 0.47 \text{ m} \tag{9}$$

The threshold value of 0.47 m is defined as the sum of the maximum and minimum radii. We chose this distance because if two particles are within less than 0.47 m of each other, it indicates recent collisions.

Fig. 6 provides an example of cluster formation in simulated bidirectional flow, illustrating the system's state at a specific simulation time of $t = 15$ s for each of the models, all beginning with the same initial conditions. Different colors represent the formed clusters, with colorless particles belonging to clusters of size one. The displayed radius corresponds to $r(t)$.

Note that in Fig. 6(c), the particles appear to overlap with the walls. This is because the collision detection criterion only considers the minimum radius, not $r(t)$. The same applies to Fig. 6(b), where the particles also appear to overlap with other particles. However, this is due to the anisotropy of the model, which only detects collisions in a certain region of the front of the particle. The only model that does not allow any overlap with the walls is the CPM. Therefore, although the particles appear to be close to the walls, there is no actual overlap. The overlaps that are observed between particles are indicators that in the next time step, the particles will move away from the collision with maximum escape velocity

We can qualitatively observe that the CPM and A-CPM form an artificial packing of particles due to the absence of avoidance mechanisms. On the other hand, the AA-CPM displays a more fluid behavior. To confirm this observation, we analyze the clusters formed by each model throughout the entire simulation and compare them with the experimental data.

Fig. 7(a) displays the distribution of cluster sizes for each model and compares them with the experimental data. While the AA-CPM occasionally shows a lower probability of producing clusters larger than those observed in the experiments, the other two models frequently generate significantly larger clusters, sometimes resulting in the formation of a single large cluster for the entire system.

Fig. 7(b) displays the average size of the largest cluster as a function of the input flow rate, with the AA-CPM closely matching the experimental data, while the other two models consistently produce larger values.

Considering the fact that clusters larger than size 7 are not detected in the experiments, we created the plot in Fig. 7(c), which shows the percentage of particles that are part of cluster larger than size 7 at some point in their trajectory. This provides insight into the number of particles that followed smooth trajectories without being involved in congestions. Once more, the AA-CPM shows values closely matching experimental data.

Another quantitative description of the system's congestion or freedom consists in calculating the temporal average of the total number of particles participating in any cluster. This is illustrated in Fig. 7(d), where the AA-CPM also succeeds in reproducing what is observed in the experimental data.

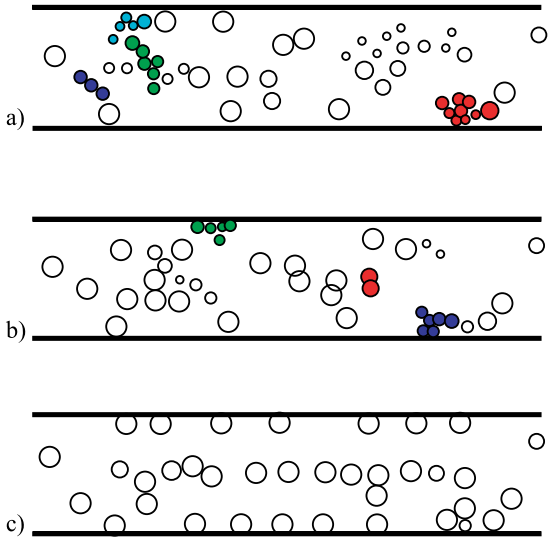


Fig. 6. Examples of cluster formation in bidirectional flow at $t = 15$ s with input flow $Q_{in} = 3.6$ 1/s and the same initial conditions simulated with: (a) CPM; (b) A-CPM; (c) AA-CPM.

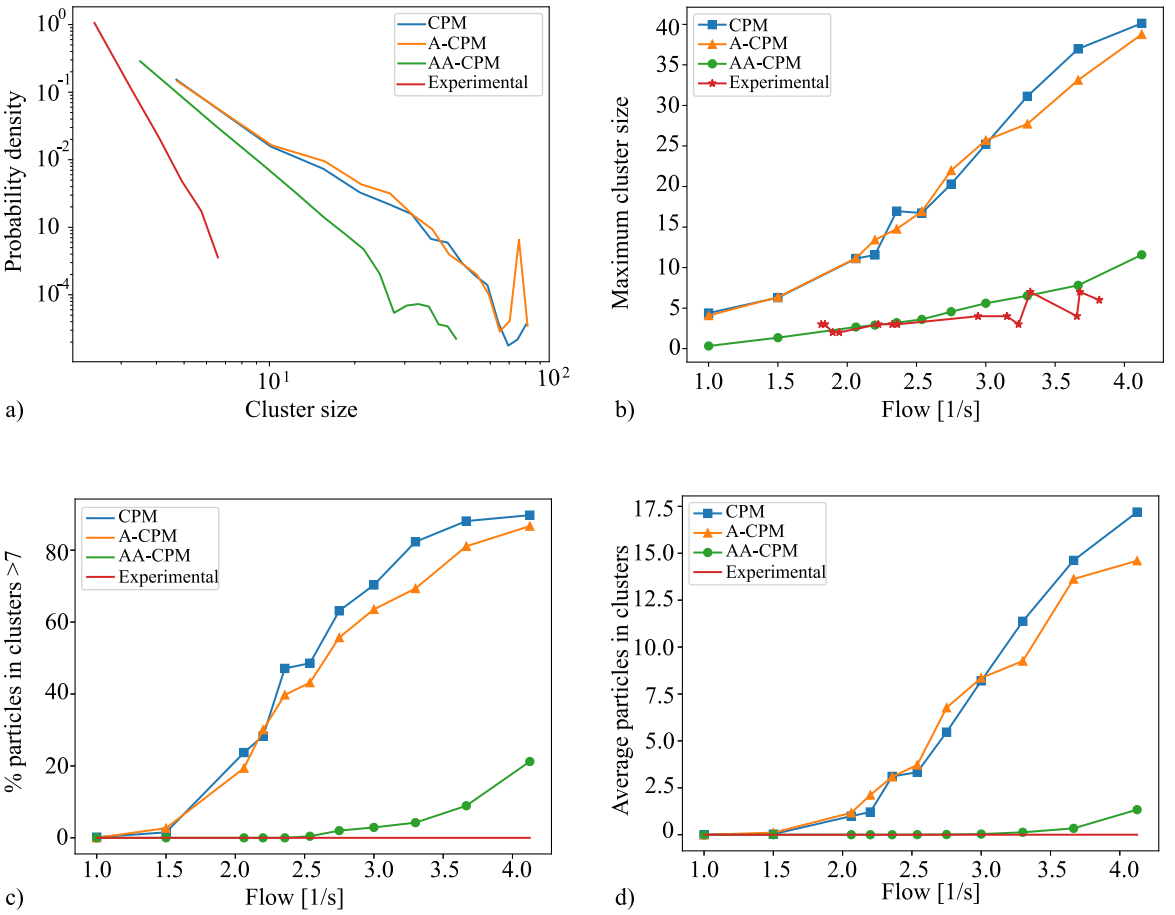


Fig. 7. Cluster analysis. (a) Cluster sizes. (b) Maximum cluster size. (c) Percentage of particles belonging to clusters with a size greater than 7 at any time. (d) Time average of the number of particles belonging to any cluster with 2 or more particles.

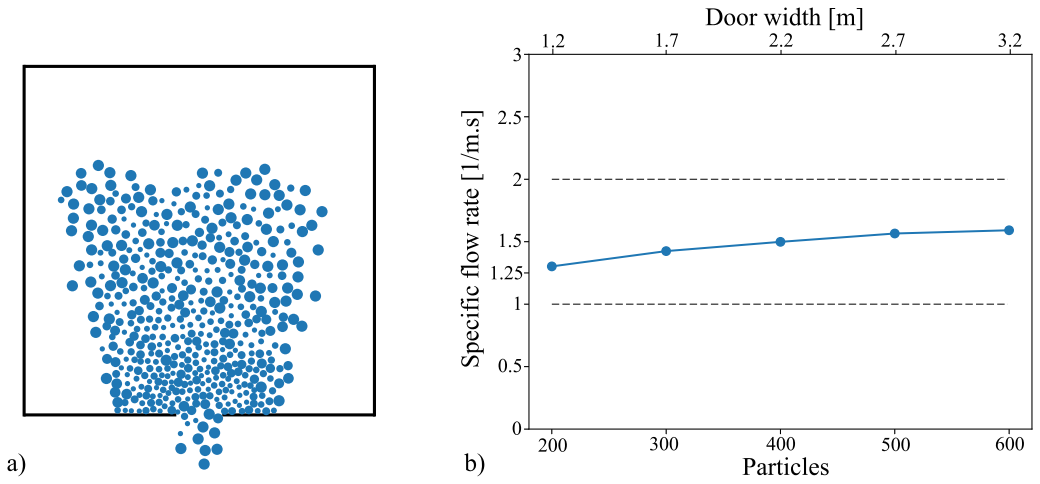


Fig. 8. Room evacuation. (a) Snapshot at time $t = 4$ s for $N = 500$ and $L = 2.7$ m. (b) Specific flow rate.

These observables clearly demonstrate the improvement achieved by the collision avoidance mechanism incorporated into the AA-CPM. It enables the simulation of bidirectional flows without artifacts, while also accurately reproducing the fundamental diagram for both unidirectional and bidirectional flows, as well as individual trajectories from avoidance experiments.

4.3. Room evacuation

After verifying the satisfactory performance of the AA-CPM in emulating experimental data from various scenarios, we aimed to investigate whether the model preserves the original CPM properties concerning the specific flow rate through doors.

To this end, we conducted simulations using the AA-CPM, in which N particles evacuate from a room of dimensions $20\ \text{m} \times 20\ \text{m}$ through an exit door located horizontally at the center of the point $(10, 0)$ with a variable width L. The target at the door for a particle with coordinates (x, y) is determined as follows:

- The point $(x, -3)$ if $y \leq 0.5$ and $10 - 0.4L \leq x \leq 10 + 0.4L$.
- The point $(x_r, 0)$ is chosen with x_r uniformly distributed between $10 - 0.4L$ and $10 + 0.4L$, otherwise.

The initial positions of the particles are randomly selected within the room with a minimum radius, and other simulation parameters are identical to those used in Section 4.1. The number of particles used is $N = \{200,\ 300,\ 400,\ 500,\ 600\}$, for values of $L = (0.2 + 0.005\ N)$ m depending on the number of particles, consistent with the method specified in [27] for calculating egress capacity. Ten realizations are performed for each N. In Fig. 8(a) we present the system state with 500 particles at 4 s after the simulation started. In this case, it is also possible to observe slight overlaps with the walls. This is due to the more relaxed collision detection criterion for walls in the AA-CPM.

The average specific flow rate for each case is calculated as $Q_e = N/(L\ T_e)$, where T_e represents the total evacuation time. Fig. 8(b) depicts the obtained results, indicating that the specific flow rate remains almost independent of the number of people, similar to the original CPM [14], with values falling within the empirical range reported in the literature [28–32] between $Q_e = 1.25\ (\text{m s})^{-1}$ and $Q_e = 2\ (\text{m s})^{-1}$.

5. Conclusions

The CPM is a recognized off-lattice cellular automaton that incorporates rules governing the movement of simulated pedestrians in continuous space. It has been shown to satisfactorily reproduce experimental unidirectional fundamental diagrams and specific flow rates [14]. In this research, we extend its capabilities through two modifications: (a) introducing Anisotropy in particle collisions, and (b) implementing Avoidance mechanisms that dynamically adjust desired velocities to prevent collisions. We refer to this new model as AA-CPM.

The parameters shared with the CPM were directly inherited. The new parameters related to the avoidance mechanism were calibrated using experimental data from pedestrian-pedestrian and pedestrian-obstacle trajectories. After fixing the parameters of the pedestrian model, it was successfully tested against experimental data from various scenarios encompassing different levels of complexity: (a) individual trajectories; (b) unidirectional flow; (c) bidirectional flow; and (d) egress through narrow doors. The study includes validation through the following observables: (1) mean distance between simulated and real trajectories; (2) fundamental diagrams; (3) cluster structure; and (4) specific flow rates through exits.

While the CPM can accurately simulate scenarios (b) and (d), the new AA-CPM not only maintains this ability but also excels in simulating scenarios (a) and (c). This is in contrast to the limitations of the original CPM, which lacked the ability to simulate these scenarios.

An important aspect of the new model is that it is not necessary to turn the avoidance mechanism on or off. When it begins to fail at high densities, the low-level A-CPM comes into play, and thus the behavior of the dense assembly matches experimental data.

As is customary in pedestrian dynamics modeling, this model accurately reproduces the observed phenomena within the examined parameter space. However, extending the application of the model to encompass additional observables or varying parameter ranges should be undertaken judiciously, requiring a rigorous validation process involving new experimental comparisons.

CRediT authorship contribution statement

Rafael F. Martin: Methodology, Software, Formal analysis, Visualization, Writing – original draft. **Daniel R. Parisi:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used chatGPT and Bard in order to improve the English writing and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgments

The authors acknowledge financial support via project PID2015-003 (Agencia Nacional de Promoción Científica *y* Tecnológica, Argentina; Instituto Tecnológico de Buenos Aires; Urbix Technologies S.A.) and ITBACyT-2018-42 (Instituto Tecnológico de Buenos Aires).

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